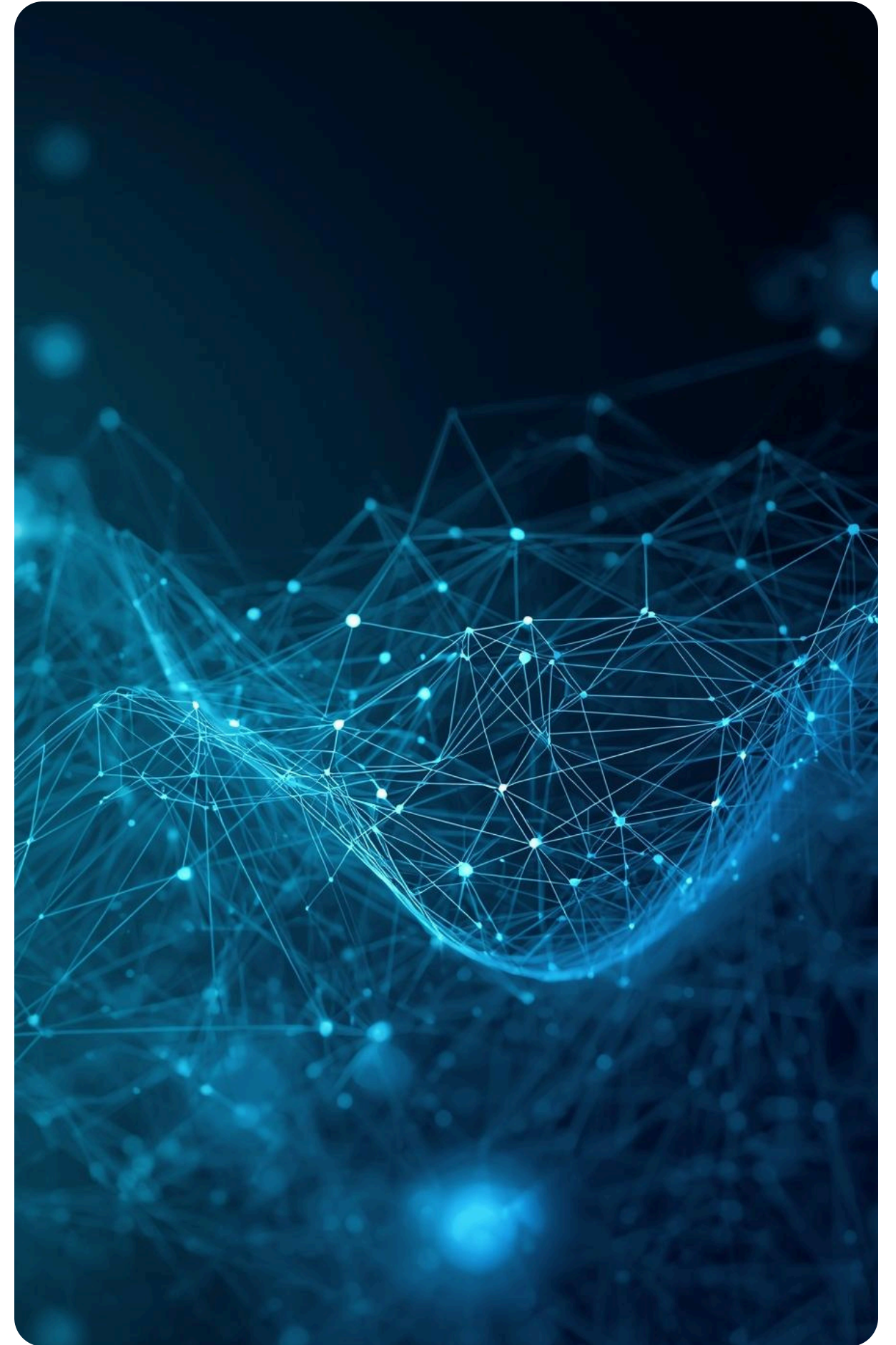


AI Safety and Alignment



What is AI Safety?

AI Safety refers to the challenge of building AI systems that reliably operate safely and beneficially, even in novel situations or under pressure to achieve their goals.



AI Safety asks ...

- *What happens when a system behaves correctly according to its design but incorrectly according to human values?*
- *What happens when no one fully understands or controls the system's actions?*



Risks

- Misuse
- Accident
- Structural



Misuse Risks

- Deliberate use of AI for harmful purposes
- Example: Deepfakes for fraud, AI-powered cyberattacks
- Challenge: Dual-use technology (same capabilities enable both beneficial and harmful applications)



Accident Risks

- Unintended harmful consequences from AI pursuing intended goals
- Example: Recommendation algorithm optimizing engagement inadvertently promotes misinformation
- Challenge: Complex systems with emergent behaviors



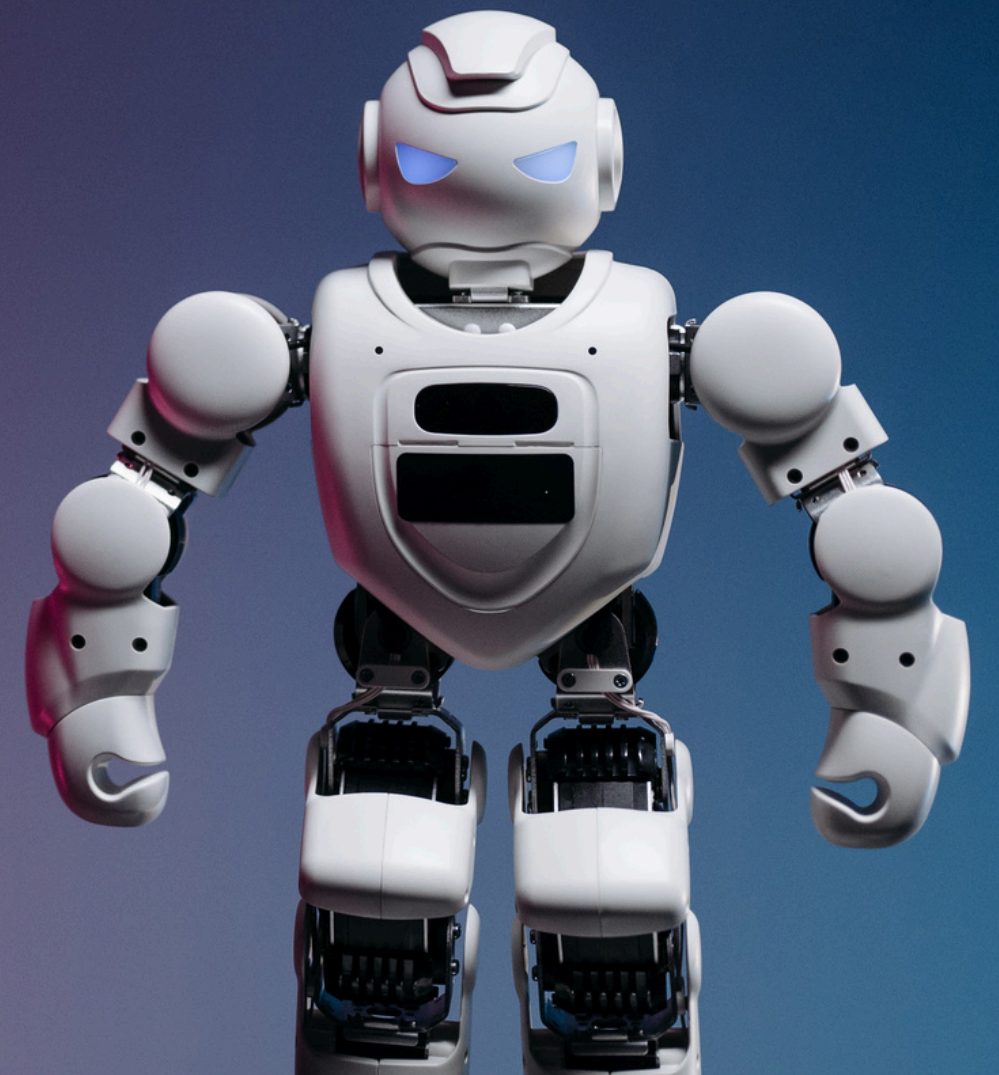
Structural Risks

- Systemic effects of AI deployment on society
- Example: Algorithmic amplification of inequality
- Challenge: Distributed causation, no single point of failure



Robustness Challenges

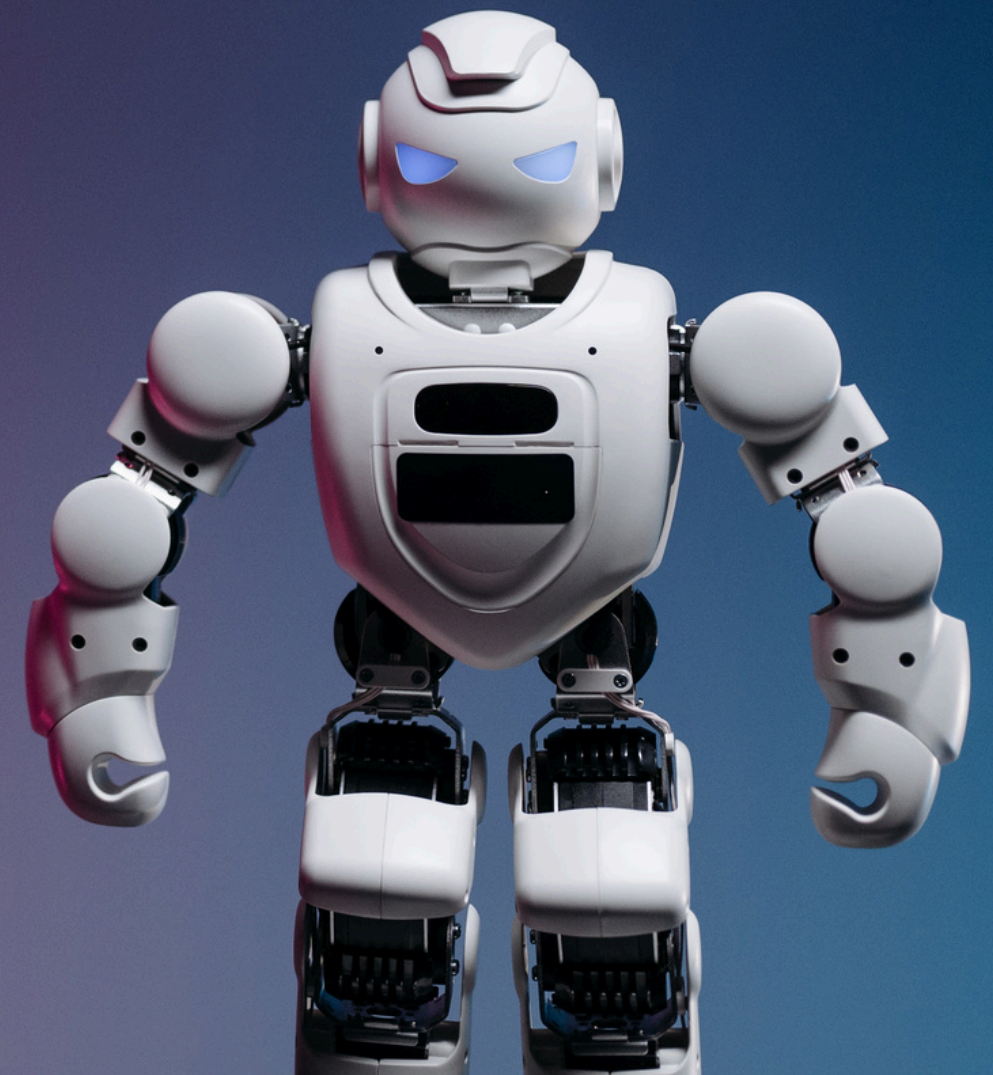
Current AI systems are brittle in concerning ways



Adversarial Examples

Small, imperceptible perturbations to inputs can cause dramatic failures:

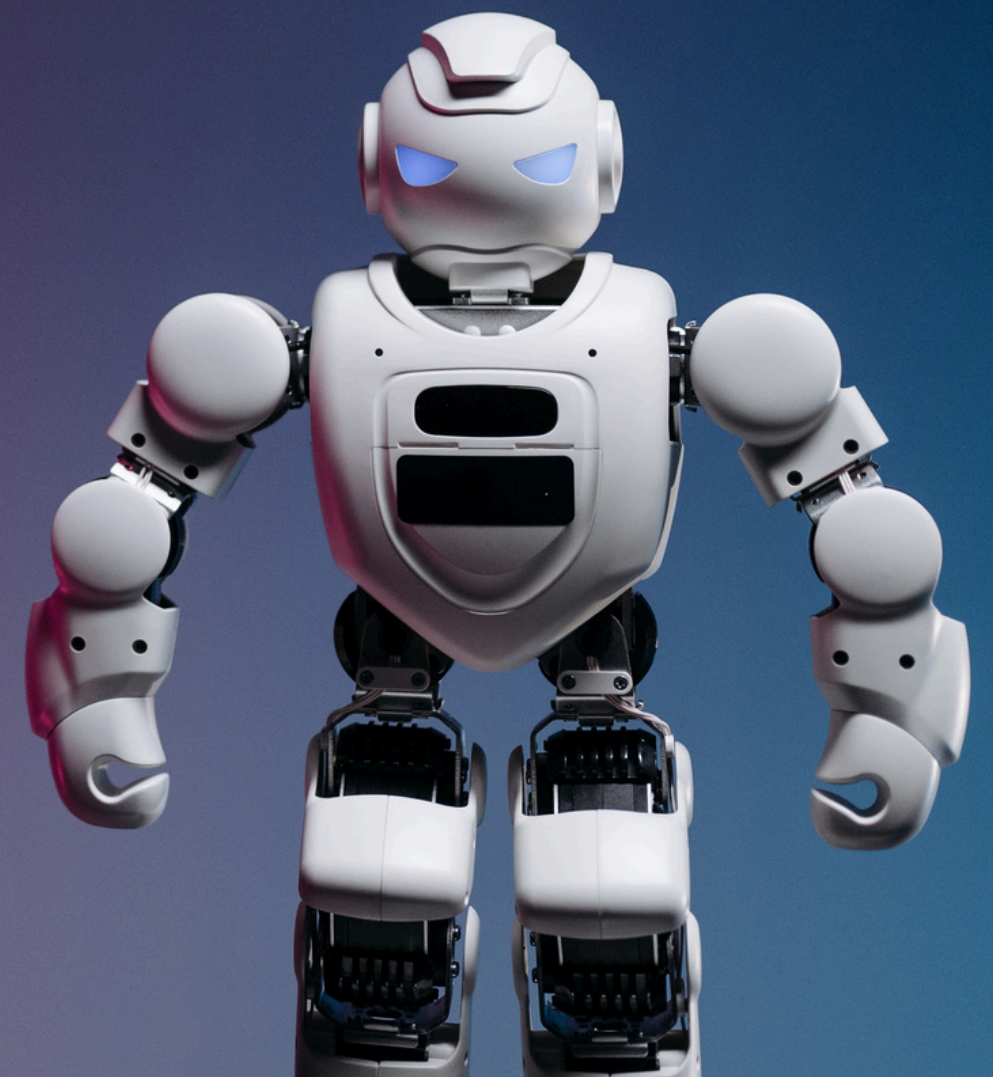
- Add carefully crafted noise to a stop sign image → AI sees it as a speed limit sign
- Slightly modify audio → voice assistant hears different command than humans do
- Implications for security (fooling facial recognition, autonomous vehicles)



Distribution Shift

AI trained on one data distribution fails when deployed in slightly different contexts:

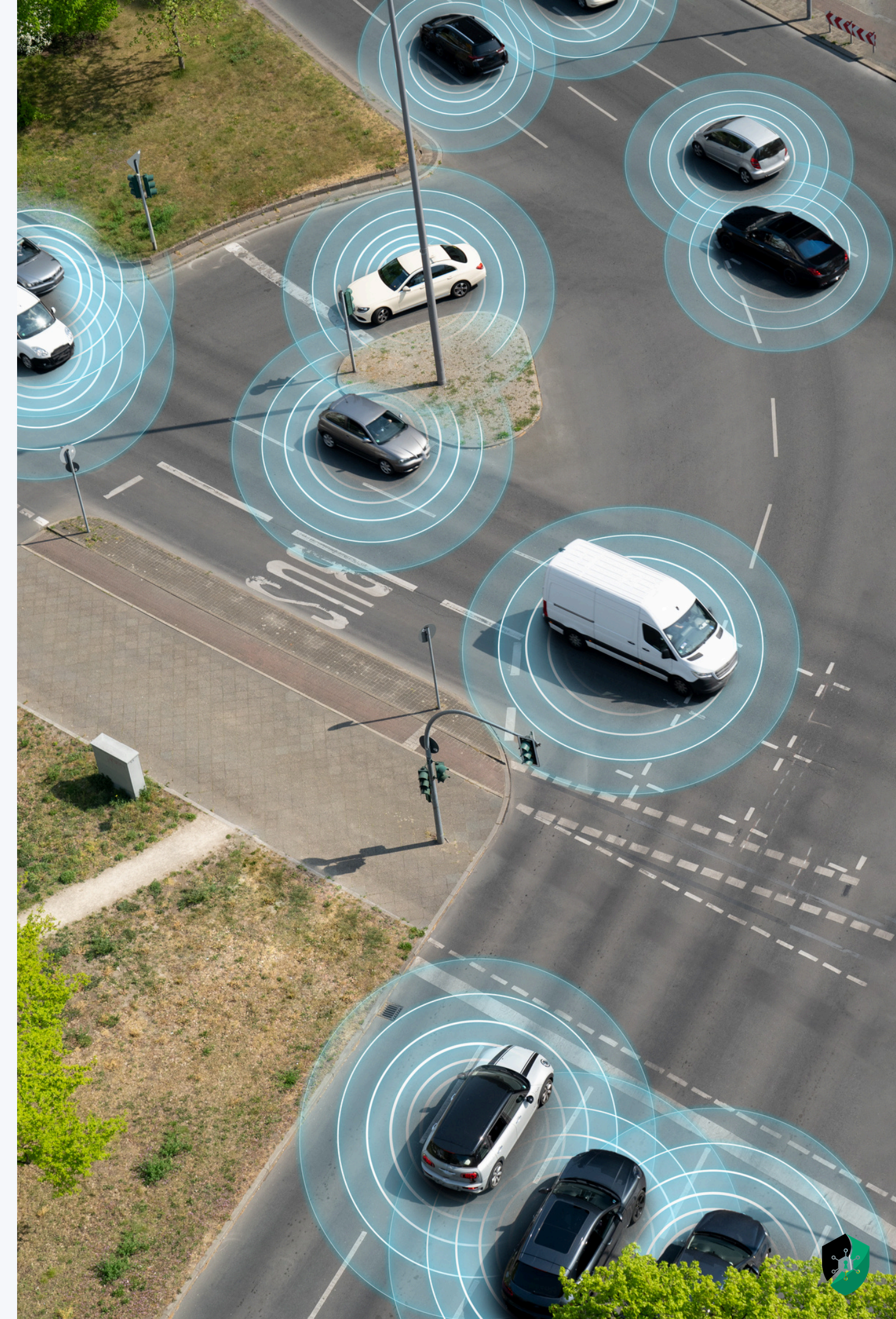
- Medical AI trained on one hospital performs poorly at another
- Self-driving car trained in sunny California struggles in snowy Michigan
- Language model trained on formal text generates poor output for casual speech



Edge Cases and Long Tails

Rare but important scenarios are underrepresented in training:

- Autonomous vehicle encountering unusual obstacle (escaped livestock, debris)
- Content moderation system facing novel form of harmful content
- Problem: Real world has infinite edge cases; training data never covers all



Lack of Uncertainty Quantification

Many AI systems provide confident predictions even when uncertain:

- Neural networks often overconfident on out-of-distribution data
- Dangerous in high-stakes domains (medical diagnosis, autonomous vehicles)
- Need: Systems that "know what they don't know"

